Conditional variational autoencoders

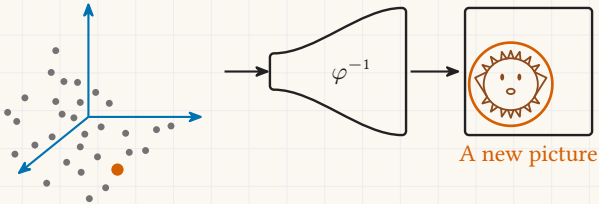
Weekend 4, Friday morning

Carlos Cotrini

CAS Building ML/AI Applications, ETH Zurich

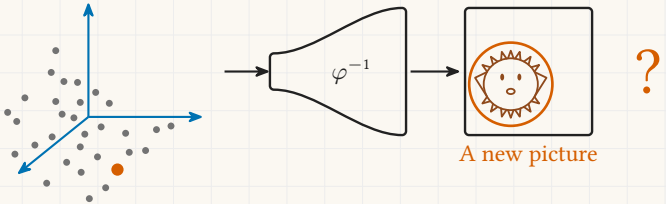
Friday 23 October 2026

Which fish will it be?



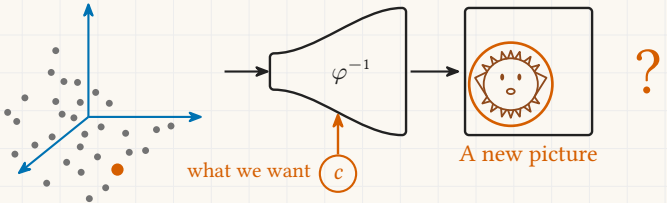
1. This morning: pick any point in the cloud of codes, decode it, and a new pufferfish appears.

Which fish will it be?



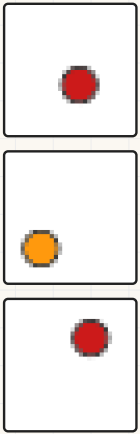
1. This morning: pick any point in the cloud of codes, decode it, and a new pufferfish appears.
2. Which one? Nothing in the cloud says where the puffed-up fish are, or the fish at the top.

Which fish will it be?

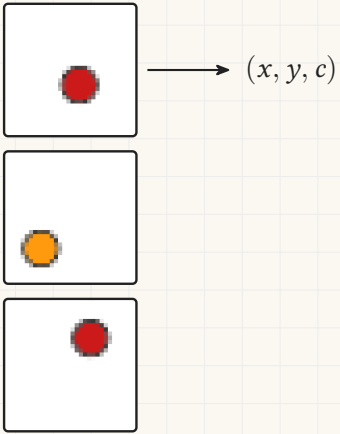


1. This morning: pick any point in the cloud of codes, decode it, and a new pufferfish appears.
2. Which one? Nothing in the cloud says where the puffed-up fish are, or the fish at the top.
3. A conditional VAE gives the decoder a second input: a label c that says what we want.

Three numbers per picture

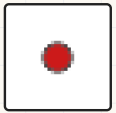


Three numbers per picture

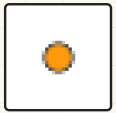


x, y: where the fruit is

c: which fruit it is, the label

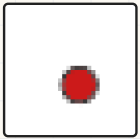


$c = 0$
an apple

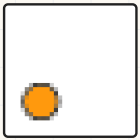


$c = 1$
an orange

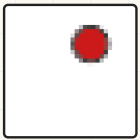
Three numbers per picture



$(x, y, c) = (0.57,\ 0.38,\ 0)$



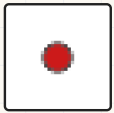
$(x, y, c) = (0.27,\ 0.26,\ 1)$



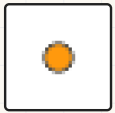
$(x, y, c) = (0.66,\ 0.71,\ 0)$

x, y: where the fruit is

c: which fruit it is, the label



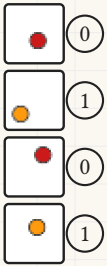
$c = 0$
an apple



$c = 1$
an orange

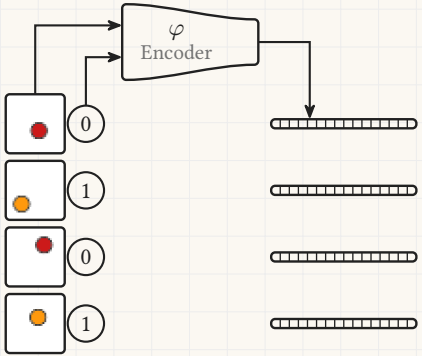
Every picture comes with its label c.

The yogurt-cup game, with labels



Pictures and labels

The yogurt-cup game, with labels

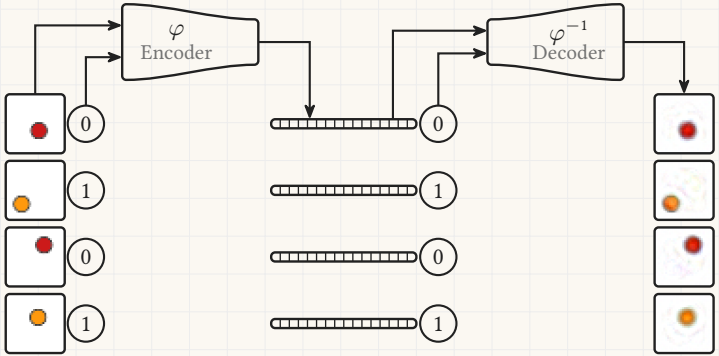


Pictures and labels

Codes

16 numbers per picture

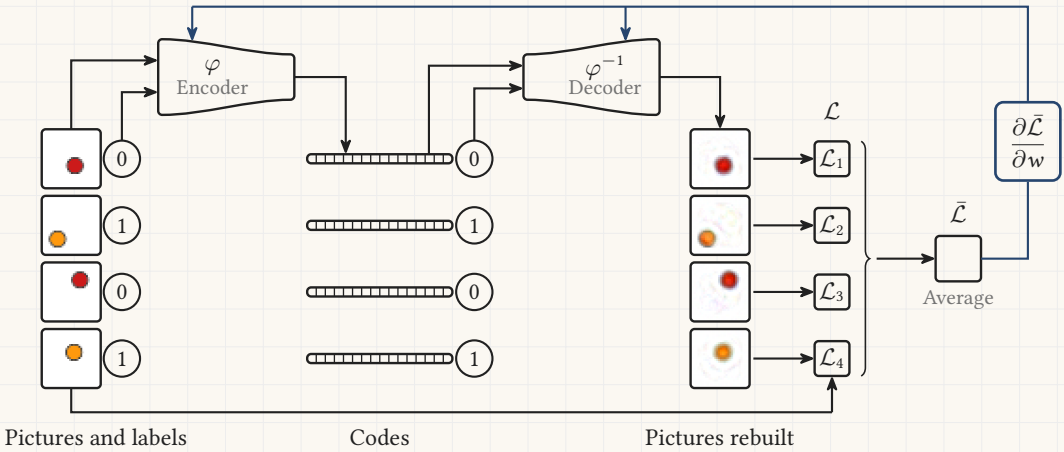
The yogurt-cup game, with labels



Pictures and labels | Codes | Pictures rebuilt

16 numbers per picture, then its label

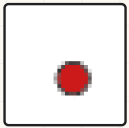
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16 numbers per picture, then its label

Flipping the label

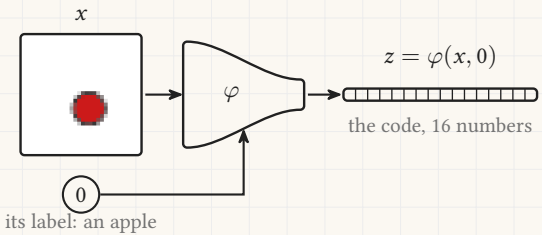
x



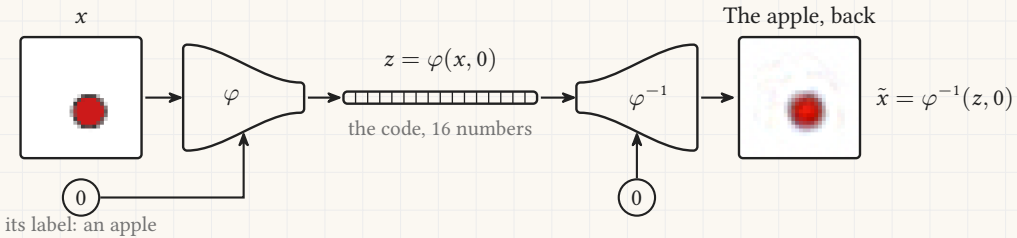
0

its label: an apple

Flipping the label

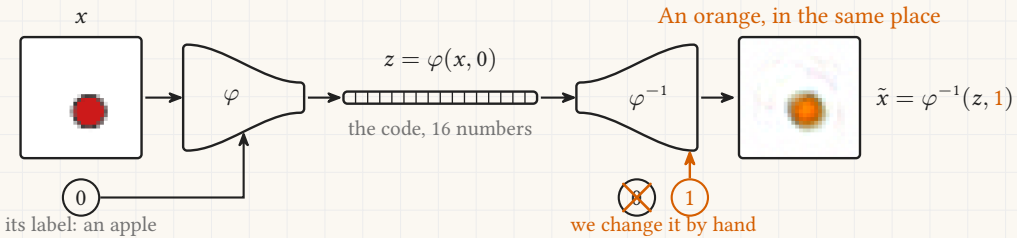


Flipping the label



1. Encode the apple with its label, decode it with the same label: the apple comes back.

Flipping the label



1. Encode the apple with its label, decode it with the same label: the apple comes back.
2. Decode the same code with the other label: an orange, in the same place.

Sixteen numbers, room to spare

Trained without a penalty, then the label of every one of the 5000 fruit flipped:



A flip worked if the picture is closer to the other fruit, drawn in the same place, than to the fruit we started from.

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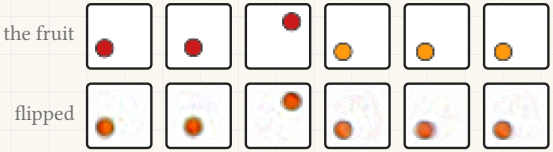
6 of the 608 that failed, chosen because they failed

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6 of the 608 that failed, chosen because they failed

1. Where a fruit is takes two numbers. The code has sixteen: room to say which fruit, too.
2. So the decoder takes the fruit partly from the code. Flip the label, and the fruit changes only part of the way: half apple, half orange.

A penalty on the length of the code

$$\mathcal{L}_i = \underbrace{\text{miss}_i}_{\text{as before}} + 0.001 \times \underbrace{(z_1^2 + z_2^2 + \cdots + z_{16}^2)}_{\text{the squared length of the code}}$$

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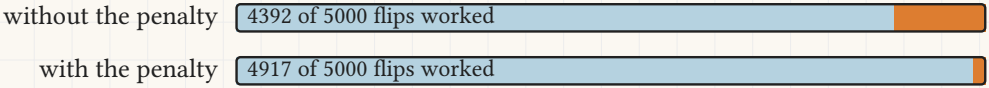
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Every number of the code now costs. The label says which fruit for free, so the code is pushed to leave the fruit to the label, and the decoder learns to take it from there.

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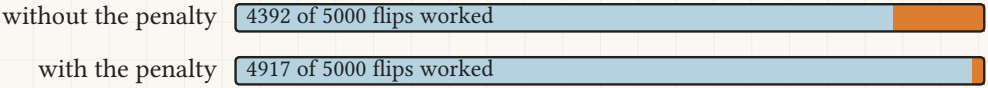


one run of the notebook, seed 0; on Colab the counts can differ a little

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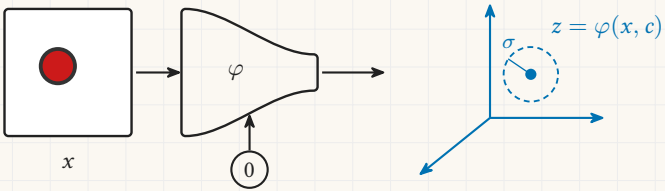
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one run of the notebook, seed 0; on Colab the counts can differ a little

The code still knows the fruit, only less: of the 10 codes nearest to each fruit's own, 90% are the same fruit, against 97% without the penalty, where half would mean it knows nothing.

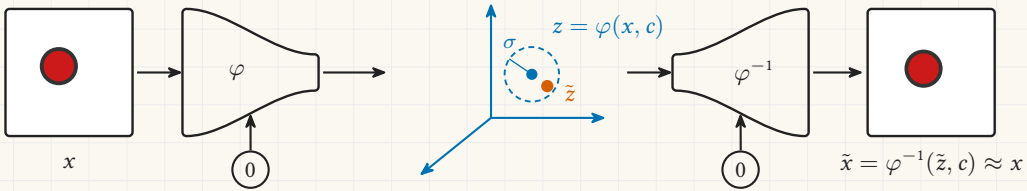
Conditional variational autoencoders



1. The encoder gets a picture x and its label c, and gives a centre $z = \varphi(x, c)$ and a radius σ.

Kingma, Rezende, Mohamed and Welling, NIPS 2014; Sohn, Lee and Yan, NIPS 2015

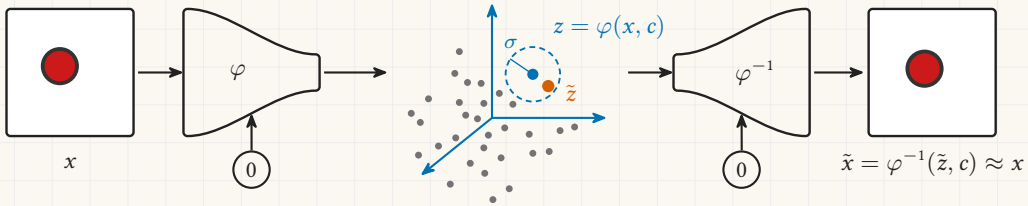
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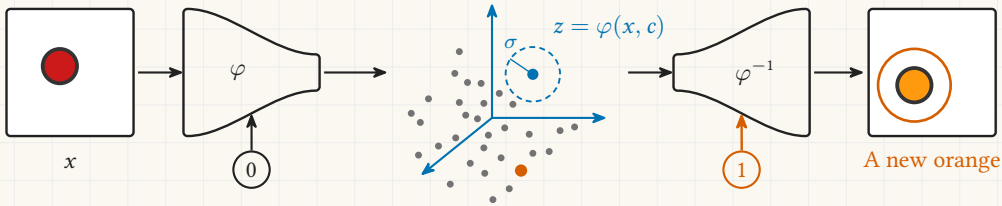
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4. To make a new picture, pick any point in the cloud, choose c, and decode it.

Kingma, Rezende, Mohamed and Welling, NIPS 2014; Sohn, Lee and Yan, NIPS 2015

Age progression

	Apples and oranges	Faces
The picture x	a fruit on white paper	
The label c	which fruit: 0 apple, 1 orange	
The code z	where the fruit is	
Keep z, change c	the same place, the other fruit	
It needs	the fruit of every picture	

Age progression

	Apples and oranges	Faces
The picture x	a fruit on white paper	a photo of a face
The label c	which fruit: 0 apple, 1 orange	the person's age, a number
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The label c	which fruit: 0 apple, 1 orange	the person's age, a number
The code z	where the fruit is	the rest: who, the pose, the light
Keep z, change c	the same place, the other fruit	the same person, at another age
It needs	the fruit of every picture	the age of each person in the photos

It can change only what the training pictures were labelled with.

The apples and oranges notebook



1. 5000 made-up fruit, each drawn from its three numbers (x, y, c).
2. A conditional autoencoder with a code of 16 numbers, trained twice: without and with the penalty.
3. It flips the label of all 5000 fruit, and counts the flips that worked with each.
4. The last cell is *Flipping the label*, live: pick a fruit, choose its label and the autoencoder.

`https://colab.research.google.com/github/eth-bmai-hs26/w4-cx-public/blob/main/apple-orange-autoencoder/apple_orange_autoencoder.ipynb`